

INFORMATION SYSTEM PROJECT DOCUMENTATION GUIDELINES

Subject: ITCC 104 – Database Management System

I. DOCUMENT FORMAT REQUIREMENTS

Page Setup

- Paper Size: A4
- Margins:
 - Left: 1.2 inches
 - Top: 1 inch
 - Right: 1 inch
 - Bottom: 1 inch

Font Specifications

- Font Style: Bookman Old Style
- Font Size:
 - Title Headings: 12 pt, Bold
 - Regular Text: 12pt

Page Numbering

- Page numbers shall be placed at the **top-right corner** of each page.
 - Chapter title pages shall not display page numbers.
 - All page numbers must correspond correctly with the Table of Contents.
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COVER PAGE

The cover page shall be designed in the form of a **Project Poster** containing the following information:

- Agusan del Sur State University
- Project Title
- Subject Code: ITCC 104
- Subject Description: Database Management System
- Date of Submission
- Name of Instructor: BERNIE S. BALIGHOT
- Academic Rank: ASSOCIATE PROFESSOR I

- Name of Student
 - Course, Year, and Section
-

TABLE OF CONTENTS

Provide a complete and properly formatted Table of Contents showing all chapters, sections, subsections, and corresponding page numbers.

CHAPTER 1 – SYSTEM OVERVIEW

A. System Overview

Provide:

- System Name
- System Description
- Purpose of the System

B. Target Users of the System

Identify and describe the intended users of the system.

Examples:

- Administrator – Description
- Staff – Description
- Manager – Description
- Customer – Description
- Guest User – Description

C. Scope of the System

Describe the functionalities available to each user role.

User Privileges

Present the scope according to the permissions granted to:

- Administrator
 - Enumerate all the functionalities that this user can perform
- Staff

- Customer
- Other identified users

D. Technology Stack

Describe the technologies used in developing the system.

Front-End Technologies

Examples:

- HTML – Description
- CSS – Description
- JavaScript – Description
- ReactJS – Description

Back-End Technologies

Examples:

- PHP – Description
- Node.js – Description
- Express.js – Description

Database Technologies

Examples:

- MySQL – Description
 - MariaDB – Description
 - PostgreSQL – Description
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CHAPTER 2 – SYSTEM ARCHITECTURE

A. System Architecture Diagram

Present diagrams showing:

- Overall system workflow
- User interaction with the system
- Server architecture
- Database interaction

Discussion

Provide a detailed explanation for each architecture diagram, including:

- Components involved
 - Functions of each component
 - Data flow between components
 - Deployment and installation setup
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CHAPTER 3 – UNIFIED MODELING LANGUAGE (UML)

A. Introduction to UML

Discuss:

- Purpose of UML
- Importance of UML in System Development
- UML diagrams used in the project

B. Use Case Diagram

Present the Use Case Diagram showing:

- Actors
- Use Cases
- Relationships

Discussion

Explain:

- Purpose of the Use Case Diagram
- Responsibilities of each Actor
- System functionalities represented by each Use Case

C. Use Case Descriptions

Provide detailed descriptions of each Use Case using a tabular format containing:

- Use Case Name
- Actor
- Description
- Preconditions

- Main Flow
 - Alternative Flow
 - Postconditions
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CHAPTER 4 – SEQUENCE DIAGRAMS

A. Sequence Diagrams

Create Sequence Diagrams for every identified Use Case.

Discussion

For each diagram, explain:

- User interactions
- System processes
- Database transactions
- Flow of information between components

Each sequence diagram must clearly identify:

- Actor
 - System Interface
 - Application Logic
 - Database
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CHAPTER 5 – DATABASE DESIGN

A. Entity Relationship Diagram (ERD)

Overview of Entities

Provide a description of all entities included in the database.

Database Schema

Present the database structure including:

- Tables
- Primary Keys
- Foreign Keys

Data Dictionary

Provide a table containing:

- Field Name
- Data Type
- Length
- Constraints
- Description

Entity Relationship Diagram

Present the complete ERD of the system.

Relationship Description

Explain:

- One-to-One Relationships
- One-to-Many Relationships
- Many-to-Many Relationships

Discuss how entities are connected and why the relationships are necessary.

CHAPTER 6 – SQL IMPLEMENTATION OF CRUD OPERATIONS

A. Data Entry (CREATE)

Data Entry Forms

Provide screenshots of all implemented data entry forms.

Source Code

Include the source code that performs:

- Form Validation
- Data Processing
- Database Insertion

Explain how data flows from: Front-End → Back-End → Database

B. Data Views (READ)

Screenshots

Provide screenshots showing the display of stored records.

Source Code

Include the code responsible for:

- Retrieving data
- Displaying records
- Formatting output

Explain how the retrieval process works.

C. Data Manipulation (UPDATE)

Screenshots

Provide screenshots showing:

- Edit Forms
- Update Processes

Source Code

Include the code responsible for:

- Editing records
- Updating records in the database

Explain the update workflow.

D. Search and Delete Records (DELETE)

Screenshots

Provide screenshots of:

- Search functionality
- Record listing
- Delete confirmation

Source Code

Include the code responsible for:

- Searching records
- Filtering results
- Deleting records from the database

Explain how the search and delete processes operate.

SUBMISSION REQUIREMENTS

Students must submit:

1. One (1) Printed and Soft-Bound Copy of the Documentation. Orange Color Tape Binding
2. One (1) Digital Copy (PDF Format).
3. Project Source Code (Softcopy).
4. SQL Database File.
5. Accomplished Rating Sheet.

Failure to comply with the documentation format and submission requirements may affect the project evaluation and grading.